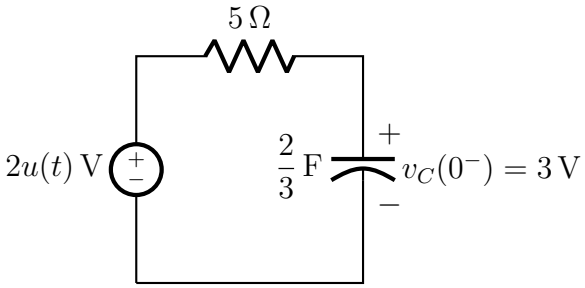
EE2015: Electric Circuits and Networks

Tutorial 1 Answer Key
(August 8, 2025)

1. (a) (0 V, 1 mA), (0 V, 1 mA), ?, (0 V, $\frac{1}{7}$ mA)
   (b) $i_C = 0$, $i_C = 0$

2. $-0.6\,\text{mW}$

3. (a) $V_c(s) = \dfrac{30s + 6}{10\,s^2 + 3s}$
   (b) $v_c(t) = (e^{-0.3t} + 2)\ u(t)$
   (c)



4. (5 V, 3 A)

5. (a) $4\,\dfrac{di_1}{dt} + 4\,i_1 = \cos(t)\,u(t)$
   $i_1(t) = \dfrac{1}{8}\left[-e^{-t} + \cos(t) + \sin(t)\right]\ u(t)$
   (b) $v_2(t) = -\dfrac{1}{20}\left[e^{-t} + \cos(t) - \sin(t)\right]\ u(t)$

6. $R_{\text{eq}} = 230\,\text{k}\Omega$